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RESEARCH ARTICLE

# Multimodal Daily-Life Emotional Recognition Using Heart Rate and Speech Data From Wearables

**EESUN MOON[1], A. S. M. SHARIFUZZAMAN SAGAR[2], (Student Member, IEEE), AND HYUNG SEOK KIM[2], (Member, IEEE)**

[1]Department of Intelligent Mechatronics Engineering, Sejong University, Seoul 05006, Republic of Korea
[2]Department of Artificial Intelligence and Robotics, Sejong University, Seoul 05006, Republic of Korea

Corresponding author: Hyung Seok Kim (hyungkim@sejong.ac.kr)

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**ABSTRACT** Human emotion plays a significant role in mental well-being, and recognizing these emotions in daily life is essential. With the advancement of artificial intelligence, affective computing has paved the way for effective applications in enhancing emotional states in everyday life. In practical daily-life scenarios, data sources that can be collected through simple and low-cost wearables contribute to recognizing emotions in daily routines. Heart rate and speech data can be easily collected from affordable smartwatches without any other human intervention. Heart rate data, directly correlated with physiological response, and speech data, with its rich expressiveness, together yield a robust indicator of the human's emotional condition. We conduct multimodal emotion recognition (MER), integrating heart rate-based emotion recognition (HER) and speech-based emotion recognition (SER) through a score-based fusion method. Our proposed MER achieves an overall accuracy of 84.22%, surpassing single-modality models with accuracies of 57.65% for HER and 80.38% for SER. The findings highlight the practicality of utilizing emotion-related data sources collected conveniently with smartwatches, thereby enhancing emotion tracking accessibility in daily life scenarios. Furthermore, integrating these modalities proves more effective in capturing emotions than using a single modality. Moreover, our system's lightweight architecture facilitates easy expansion to incorporate additional modalities, ensuring durable precision even when not all modalities are sensed, making it versatile and pragmatic.

**INDEX TERMS** Affective computing, deep learning, emotion recognition, fusion, multimodal.

## I. INTRODUCTION

Emotions, both transient and sustained, significantly influence our mental health. The ability to detect these emotional cues holds transformative potential for personalized mental health care. Affective computing primarily focuses on deciphering, understanding, and responding to human emotions, aiming to bridge the gap between human emotional states and technological applications. The advancement of Artificial Intelligence has led to applicable fields in affective computing, leveraging emotion recognition, human-robot interaction, and multimodal signal processing for a holistic understanding of human emotional states [1].

Human emotions are expressed through diverse methods, including facial expressions, text, speech, body gestures, and physiological responses. Heart rate (HR) variations reflect the degree of change in emotional state, closely linked to the human autonomic nervous system [2]. In essence, HR is involuntary and cannot be consciously controlled by individuals [3]. Speech plays a crucial role in human emotional expression, with non-verbal features like pitch, tone, speed, and volume serving as vital cues for understanding emotional states [4]. Both HR and speech data can be conveniently collected using simple and low-cost wearable sensors like smartwatches, making them suitable for practical emotion

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management scenarios in everyday life without complex equipment or movement constraints.

While previous studies have made significant strides in emotion recognition, they often face limitations stemming from a singular focus on specific data sources. Whether analyzing facial expressions, speech, or physiological signals, these approaches may not fully capture the full spectrum of human emotions, leading to reduced accuracy and applicability in diverse environments. For instance, Mellouk and Handouzi achieved moderate classification accuracies using photoplethysmography (PPG) derived from facial videos [5], while Fan et al. enhanced electrocardiogram (ECG)-based emotion recognition with attention mechanisms [6], yet faced challenges in maintaining high accuracy across different datasets. Similarly, Liu et al. and Kumar et al. reported varied success rates in speech emotion recognition, which, although promising, highlight the variability and the need for more robust solutions [7], [8].

Emotion recognition involves leveraging multiple modalities to interpret and pinpoint human emotional states. By fusing various emotion-related data sources, researchers can better grasp human emotions and enhance the precision of emotion analysis [9]. Specifically, while HR data obtained through physiological sensing can be continuously collected over time, it may suffer from significant noise during the data acquisition process. Conversely, speech data, though more intuitive and robust for emotion recognition, can be challenging to obtain continuously as individuals do not speak at every point. Multimodal fusion techniques can address these limitations by harnessing the combined strength of diverse data streams, augmenting data richness as well as reducing uncertainties. For example, recent efforts, including those by Le et al. and Liu et al., introduced systems that integrate audio, visual, and textual signals, achieving notable improvements in emotion recognition accuracies [10], [11]. Moreover, Jiang et al. and Zhang et al. utilized advanced architectures like fuzzy logic and hybrid attention networks to refine the accuracy and responsiveness of their models [12], [13]. Although numerous studies have been conducted to improve the accuracy of emotion recognition, studies focusing on emotion recognition in complex and variable daily life scenarios remain limited. Existing approaches often overlook the practical hurdles associated with real-world applications, such as the diverse contexts and dynamic nature of human emotions especially encountered in daily life.

Previous works often rely on single modality approaches or complex computational models, which may not be suitable for lightweight and portable systems required for practical deployment in real-world settings. Therefore, the main focus of this study is to enhance emotion recognition by integrating multiple modalities and to develop a practical and efficient multimodal emotion recognition system suitable for dynamic real-world applications. Leveraging the characteristics of wearable devices, such as the simplicity and convenience of low-cost smartwatches, and utilizing HR and speech data easily obtainable from them, our goal is to achieve relatively high accuracy in emotion recognition while ensuring the system remains lightweight and easily deployable in diverse scenarios.

By utilizing the continuous physiological data provided by HR and the expressive richness of speech, our system aims to provide a more accurate and reliable analysis of emotional states. Furthermore, through the fusion of these modalities using a score-based fusion approach, which reduces computational complexity by eliminating the need for additional neural networks, our system aims to furnish a comprehensive understanding of human emotional states, while also retaining flexibility for future modality extensions. This study makes a twofold contribution: Firstly, it analyzes users' emotional states in daily life using HR-based emotion recognition (HER) and speech-based emotion recognition (SER). Secondly, it achieves a comprehensive understanding of emotions through multimodal emotion recognition (MER) by integrating the HER and the SER using a score-based fusion approach.

The rest of the paper is organized as follows. In the following section, relevant literature is discussed thoroughly. In Chapter 3, we propose the MER system, comprising the HER, the SER, and the multimodal fusion approach. Chapter 4 presents the results and analysis of HER, SER, and MER, respectively. In Chapter 5, we compare our model to state-of-the-art ones, discussing its merits and demerits. Lastly, Chapter 6 discusses the findings and concludes.

## II. RELATED WORKS

Emotion recognition within the arena of natural language processing (NLP) has increasingly become a focal point of research, largely attributed to its expansive applications, including emotion understanding systems, opinion mining, and emotion generation. A review of the literature revealed extensive investigations in this domain, with contributions from scholars like Sharma and Dhall [14], Sebe et al. [15], and numerous others spanning from 2014 to 2021. These studies have built upon each other, progressively illuminating the nuances of emotion recognition.

Several studies in multimodal emotion recognition have taken various innovative paths. Wang et al.'s integration of electroencephalography (EEG) and photoplethysmography with speech enhanced recognition accuracy, particularly in noisy environments [16]. Pan et al. similarly integrated facial, speech, and EEG data for improved emotion recognition [17]. Ali and Stephan's model combined ECG and HR variability with speech for emotion prediction, showing promise in the Remote Collaborative and Affective Interactions (RECOLA) database [18]. Dang et al. preprocessed heartbeat data from millimeter-wave radar with facial expressions, achieving high accuracy in both person-dependent and -independent experiments [19]. Yang et al. proposed an innovative system powered by an attention-based long-term short memory (LSTM) that harnesses a suite of smartphone and wristband sensors to pinpoint users' emotional states with precision [20].

Moving to speech-based emotion recognition, Sharafi et al. developed a multimodal neural network merging audio and visual data with impressive results on Surrey Audio-Visual Expressed Emotion (SAVEE), Ryerson Audio-Visual Dataset of Emotional Speech and Song (RAVDESS), and RDF Mapping Language (RML) databases [21]. Siddiqui and Javaid presented a dual-layered system fusing visual and auditory data [22], and Ouzar et al. employed imaging photoplethysmography to extract facial and physiological cues from videos [23]. Kumar et al. used IoT with a deep learning model for emotion recognition, combining visual and psychophysiological signals [24]. Zhou and Liu's study of facial expressions and speech for emotion recognition aided in adaptive higher education management [25], while Dilnoza et al.'s attention-based fusion method selected key facial and speech features, demonstrating superior performance on Interactive Emotional Dyadic Motion Capture (IEMOCAP) and CMU Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI) datasets [26]. Liu et al. addressed modality conflicts with an attention mechanism that improved semantic consistency across facial, speech, and text features, yielding an effective emotion recognition model [27].

Mamieva et al. proposed a multimodal emotion recognition system using an attention-based fusion of facial and speech features, which enhanced detection accuracy [26]. However, the system faced challenges with computational complexity. Kumar et al. presented a multimodal emotion recognition system that integrates speech and image data using a hybrid fusion technique, achieving 83.29% [28]. However, image acquisition was not possible as a continuous emotion recognition characteristic, which limited its usage in real-time emotion recognition techniques. Takir et al. investigated multimodal emotion recognition for hearing-impaired children by combining physiological signals and facial expressions, achieving an accuracy of 74.96% with multimodal fusion [29]. However, their study relied on specific child-oriented datasets which might not generalize well across broader populations. Sharam et al. proposed a multimodal emotional health detection system that integrated visual, audio, and textual data using fine-tuned transfer learning networks to classify emotions in real time [30]. Hou et al. developed a multimodal emotion recognition framework employing discriminative correlation fusion with a temporal alignment mechanism, achieving enhanced classification performance by addressing temporal misalignments between physiological signals [31]. However, the model might suffer from increased computational complexity due to its sophisticated temporal alignment and discriminative correlation computations.

Despite notable advancements in multimodal emotion recognition, current methodologies frequently encounter challenges such as reconciling conflicts between different modalities and effectively combining these different modalities. Addressing these complexities is crucial for developing systems that can accurately reflect the complexity of human emotional expression. The fusion of HR and speech data holds promise for solving modality conflict and intermodal relationship challenges by leveraging the complementary nature of auditory and physiological signals to provide a more holistic and accurate assessment of emotional states. Therefore, this study utilized HR and speech modalities to propose robust emotion recognition.

## III. MULTIMODAL EMOTION RECOGNITION

### A. OVERVIEW

#### 1) SCENARIO

This section provides a detailed explanation of the specific implementation methods for the MER, which comprises the HER, the SER, and multimodal fusion steps.

In this study, we set a practical scenario in daily life where smartwatches assess the user's emotional state, as depicted in Figure 1. Utilizing simple and low-cost smartwatches for data collection, we use HR and speech data sources for emotion analysis. Our primary goal is to detect the user's emotional tendencies in everyday life. Therefore, we focus on emotional states based on the emotional quadrant, rather than classifying emotions in detail, to enhance the precision of emotion recognition studies. To achieve this, we adopt one of the psychological emotion models, Russell's Circumplex Model shown in Figure 2 [32], and categorize emotions into four based on valence (activation and deactivation) and arousal (pleasant and unpleasant): high valence high arousal (HVHA), low valence high arousal (LVHA), low valence low arousal (LVLA), and low valence high arousal (LVHA).

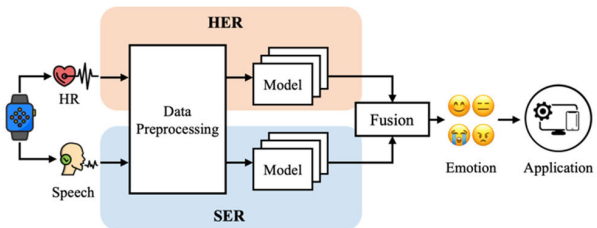


**FIGURE 1.** The framework of proposed multimodal daily-life emotional recognition using heart rate and speech data from wearables.

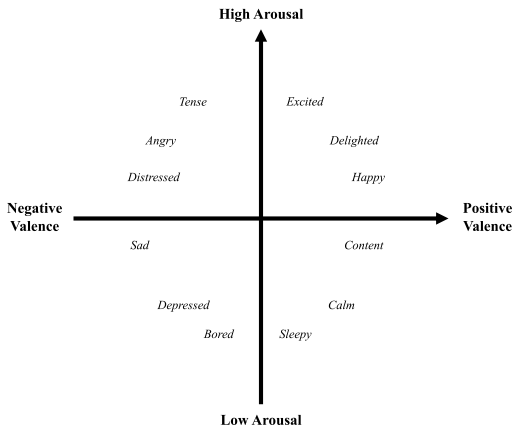


**FIGURE 2.** Rusell's Circumplex models.

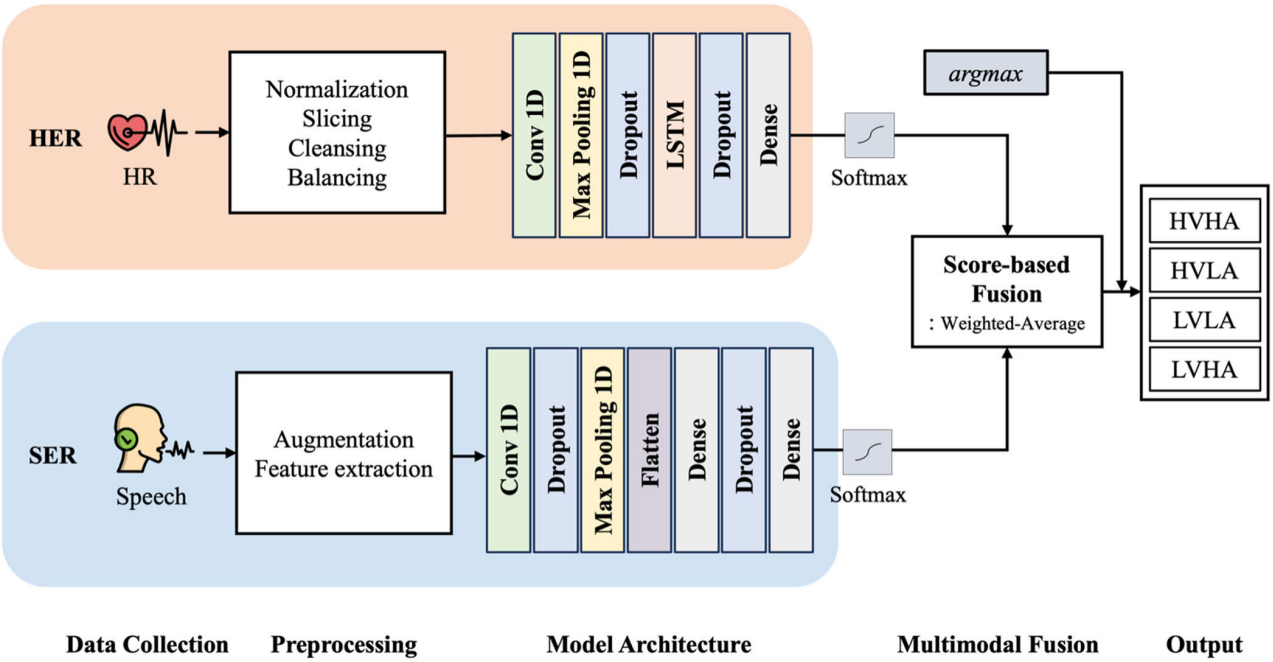


**FIGURE 3.** The overall architecture of the proposed multimodal emotion recognition (MER) model.

Our system holds potential for various applications aimed at monitoring and managing users' emotions and enhancing their emotional states in daily routines.

Based on this scenario, the proposed MER system structure is outlined in Figure 3. It consists of two main components: emotion recognition (HER and SER) and modalities' integration. The HER and the SER processes involve dataset preparation, preprocessing, and model architecture.

For the HER, we utilized the datasets, detailed in the following section, and preprocessed them, including normalization, slicing, cleansing, and balancing, in order to address the temporal characteristics of HR data. In the SER, publicly available multimodal datasets were employed due to the limited availability of datasets combining HR and speech. Preprocessing methods, including data augmentation and feature extraction, were employed to consider various audio environments and extract emotional signals from audio.

Considering our scenario embedded as a smartwatch application to monitor user emotions on a daily basis, the emotion recognition model should occupy minimal memory or capacity to efficiently operate alongside other smartwatch functions. Recent advancements in neural network research have led to the development of more efficient and sophisticated architectures [33], [34], [35], [36]. Consequently, in configuring the model architecture, we utilized relatively shallow models such as 1D convolutional neural networks (CNNs) and LSTM to effectively capture the temporal dynamics of HR and speech and extract features from each modality. The optimal configurations of emotion recognition models are determined experimentally through ablation studies, detailed in Section IV.

In the integration of modalities, since we used individual HR and speech datasets in our study, instances of HR and speech are grouped under the same emotional label. This HR-speech group is divided into train, validate, and test sets: the train set is used for model training, the validate set prevents overfitting, and the test set evaluates the trained model's performance. The HER and the SER output softmax values as predictions for each modality. These predictions are combined through the score-based fusion to derive the final prediction (HVHA, HVLA, LVLA, or LVHA) using argmax. Specifically, in the score-based fusion, the average method integrates each softmax using a weighted average based on the models' performances.

### 2) MATHEMATICAL FRAMEWORK OF THE PROPOSED EMOTION RECOGNITION MODEL

This section provides a comprehensive mathematical background for the proposed emotion recognition model, which combines 1D CNNs with LSTM networks. Our model is proposed to extract the temporal dynamics present in heart rate and speech data to accurately recognize emotional states.

- Convolutional Layer

The first layer in our architecture is a one-dimensional convolutional layer (conv1D), designed to perform feature extraction from sequential data. For an input signal $x(t)$, the convolutional operation with a kernel $k(t)$ is defined as

$$y(t) = (x * k)(t) = \sum_{i=0}^{N-1} x(i) \cdot k(t-i), \qquad (1)$$

where $y(t)$ represents the output feature map and $N$ denotes the kernel size. This process captures local temporal features essential for the initial recognition of emotional cues.

- Max Pooling Layer

The max pooling layer (max_pooling1D) is added after the conv1D layer to down-sample the feature representation, reducing its dimensionality while preserving the most salient information, which can be defined as

$$y_j = \max_{i \in [1,M]} x_{i+jM}, \tag{2}$$

where $M$ is the pooling window size, and the operation retains the maximum value within each window, contributing to the model's translational invariance and robustness.

- Dropout Layer

To mitigate the risk of overfitting, a dropout layer (dropout) randomly sets a portion of the input units to zero during training, which can be mathematically represented as

$$y = x \odot r, \tag{3}$$

where $r$ is a binary mask vector where each entry is drawn from a Bernoulli distribution with probability $1-d$, and $d$ is the dropout rate.

- LSTM Layer

The LSTM layer is introduced to capture long-range dependencies and model sequential data over extended time periods. The LSTM cell updates are governed by the following equations:

$$\text{Forget gate}: f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right), \tag{4}$$

$$\text{Input gate}: i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right), \tag{5}$$

$$\text{Cell state}: C_t = f_t * C_{t-1} + i_t * tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right), \tag{6}$$

$$\text{Output gate}: o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right), \tag{7}$$

$$\text{Hidden state}: h_t = o_t * tanh\left(C_t\right). \tag{8}$$

- Dense Layer

The architecture concludes with a dense layer (dense), which integrates the features learned by previous layers into a final prediction. The operation of the dense layer is given by

$$y = \phi\left(W \cdot x + b\right), \tag{9}$$

where $\phi$ is an activation function appropriate for the tasks, such as the softmax function for classification, $W$ represents the weight matrix, and $b$ is the bias vector.

The proposed model's architecture is mathematically constructed to integrate the strengths of CNNs and LSTMs, leveraging both feature extraction and sequence modeling capabilities. The convolutional layers effectively capture the temporal features within the data, while the LSTM layers model the sequential dependencies, essential for the nuanced task of emotion recognition. This synergy enables the proposed model to process and analyze the temporal and sequential complexities of multimodal data for robust and accurate emotion recognition.

### B. HEART RATE-BASED EMOTION RECOGNITION

#### 1) DATASET

For the HER, we utilized the dataset from previous studies, such as Tourist Experiences Recommender System based on Emotion Recognition with Wearable Data (TERS-ER) and Ulm-Trier Social Stress (Ulm-TSST) [37], [38].

TERS-ER dataset consists of data recorded by participants aged between 18 and 44 who wore a wearable wristband (Xiaomi Mi Band) and reported their daily life emotions. Participants' HR data were collected through the smartwatch, and emotion labels, including "neutral" and 17 other emotions, were directly entered by the participants through a mobile application. These emotion labels were relabeled into four categories according to the valence and arousal (VA) four-quadrant model to align with the purpose of this study, such as HVHA, LVHA, LVLA, and LVHA. Ulm-TSST is a multimodal dataset annotated with VA signals. This dataset was collected from 105 participants aged between 18 and 39 who performed oral presentations in a job interview setting, during which physiological indicators such as electrocardiography (ECG), electrodermal activity (EDA), respiration, and HR were recorded. For our study, we relabeled the continuous numerical data from this dataset into four emotion categories based on the signs of VA.

#### 2) DATA PREPROCESSING

Physiological signal data, recorded continuously over time and exhibiting unique patterns for individuals, are susceptible to potential noise during data collection. Therefore, when conducting emotion recognition research using physiological signals, these factors must be taken into consideration. For our HER using TERS-ER and Ulm-TSST, hence, we performed preprocessing steps, including participant-specific min-max normalization, data slicing considering the temporal characteristics, outlier removal to eliminate noise, and oversampling to address a class imbalance between data categories.

In participant-specific min-max normalization, physiological signal data exhibit distinct individual characteristics, which can result in variations in HR ranges among participants. These differences can lead to problems of inconsistency during model training and make it challenging to establish the relationship between HR and emotions. Therefore, to normalize HR for each participant into the same range, we applied participant-specific min-max normalization as described in

$$x_{normalized} = \frac{x - x_{min}}{x_{max} - x_{min}}, \tag{10}$$

where $x$ is the measurement of a user's heart rate, $x_{min}$ represents the minimum measurement of a user's heart rate, and $x_{max}$ denotes the maximum measurement of a user's heart rate. The participant-specific min-max normalization to HR data ensures the uniformity of scale across different individuals, mitigating the impact of inter-individual variability on the model's performance. By rescaling the

HR measurements into a comparable range, the normalization process enhances the model's ability to detect patterns and relationships between HR and emotional states more accurately.

Subsequently, we performed a slicing process taking into consideration the characteristics of time-series data. When considering the data scenario of the prior research [37], which involves participants directly inputting their emotions as experienced in daily life, we assumed that participants would have entered their emotions into the application shortly after experiencing them. Therefore, we considered the data from 15 seconds before the moment the emotion was reported to 15 seconds after, creating HR instances. The time intervals and instance sizes used for slicing were selected experimentally, and more details can be found in the Ablation Study 6 of Table 1.

Even after participant-specific min-max normalization, data may still contain noise. Therefore, in the process of data cleansing, to identify and handle outliers, we used the statistical method called the interquartile range (IQR) based on variance. We calculated the lower 25% point (quartile 1: Q1) and the upper 25% point (quartile 3: Q3) based on the variance of all participants. Subsequently, we computed the IQR, which is the difference between Q3 and Q1. Then, values smaller than the minimum (Q1 - 1.5 $\times$ IQR) or larger than the maximum (Q3 + 1.5 $\times$ IQR) were considered outliers and removed.

Class imbalance can lead to problems like overfitting and bias in model performance during training. HR data utilized in this study, such as TERS-ER and Ulm-TSST, exhibits imbalanced distributions across VA classes, as illustrated in Figure 4. Therefore, we aimed to address class imbalance using oversampling. In the oversampling process, we considered modules such as the synthetic minority over-sampling technique (SMOTE), KMeansSMOTE, and RandomOverSampler. SMOTE is a technique that improves data imbalance by synthesizing samples from the minority class based on the distance between samples in the minority class. KMeansSMOTE is a variation of SMOTE that uses the clustering algorithm K-Means for oversampling. RandomOverSampler balances the data by randomly replicating samples from the minority class. We selected the KMeansSMOTE approach through experimental methods, and this information is detailed in the Ablation Study 7 of Table 1.

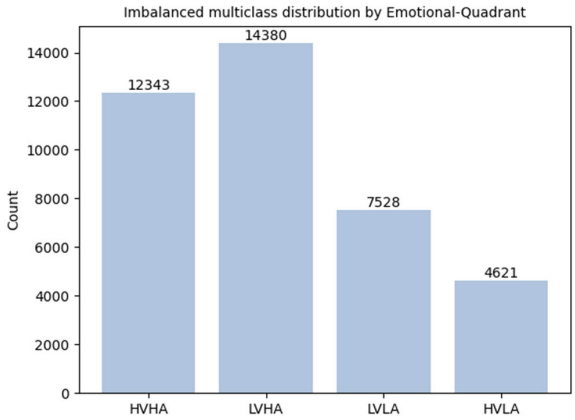


**FIGURE 4.** The amount of data by VA class.

### 3) MODEL ARCHITECTURE

Based on the preprocessed data, we constructed and trained a 1D convolutional neural network (CNN) -LSTM model [37]. The 1D CNN is highly effective at extracting temporal patterns and features from time-series data, making it well-suited for utilizing dynamic changes in HR as essential emotional features. Additionally, the LSTM, a type of recurrent neural network (RNN), is specialized in handling time-series and sequential data, making it a suitable architecture for modeling with HR data. We built three different models: 1D CNN-Flatten, 1D CNN-LSTM, and deep convolutional neural network (DCNN), and compared their performance to decide the optimal configurations of the HER. Through experimental methods represented in Ablation Study 1 of Table 1, we selected the 1D CNN-LSTM model as our final model. Figure 5 provides a visual representation of the architecture of the HER.

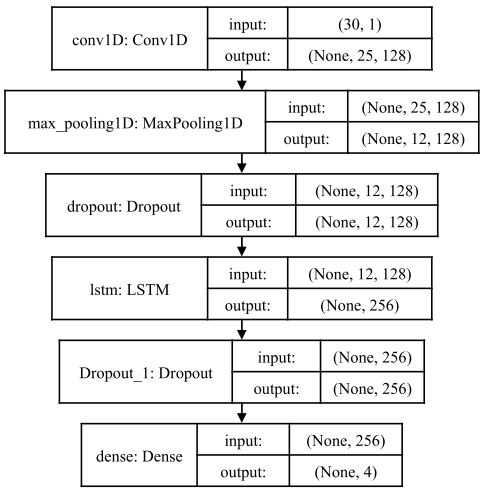


**FIGURE 5.** Block diagram of the HER: 1D CNN–LSTM.

## C. SPEECH-BASED EMOTION RECOGNITION

### 1) DATASET

For the SER, we utilized publicly available multimodal datasets, including Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D), Toronto Emotional Speech Set (TESS), RAVDESS, and SAVEE [39], [40], [41], [42].

CREMA-D consists of data from 91 actors aged 20 to 74, who selected one of six emotions (happy, sad, angry, disgusted, fearful, and neutral) and one of four emotional intensity levels (low, medium, high, and unspecified) to speak 12 sentences. TESS is described by two female actors aged 26 and 64, expressing seven emotions (happy, sad, angry, disgusted, fearful, surprised, and neutral). RAVDESS comprises audio-video data of 12 male and 12 female professional actors

speaking in seven emotional states (happy, calm, sad, angry, disgusted, fearful, and surprised). Lastly, SAVEE features recordings from four male graduate students and researchers expressing emotions in seven categories (happy, sad, angry, disgusted, fearful, surprised, and neutral).

#### 2) DATA PREPROCESSING

Audio signals are prone to contamination by surrounding noise during the data collection process. Therefore, when performing emotion recognition using speech data, it is essential to consider noise and various environmental factors while extracting crucial features from the audio signal. Consequently, to ensure the robustness of our approach, we conducted data preprocessing steps to address potential challenges inherent in audio data. Our preprocessing pipeline involved data augmentation and feature extraction processes.

Data augmentation for audio involves transforming or extending the original audio signal to increase the dataset's size. This is done to improve the model's generalization performance by considering a variety of speaking environments. Before performing the data augmentation process, we visualized the amplitude over time for different emotion classes of the original audio data, as shown in Figure 6. Based on these graphs, we set the offset to 0.6 seconds and the duration to 2.5 seconds when loading the audio data. Subsequently, to enhance the model's ability to analyze emotional nuances related to variations in pitch and improve its resilience to environmental noise, we augmented the original data by adding noise and applying pitch shifts (pitching) to the data and added noise to the pitched data as well.

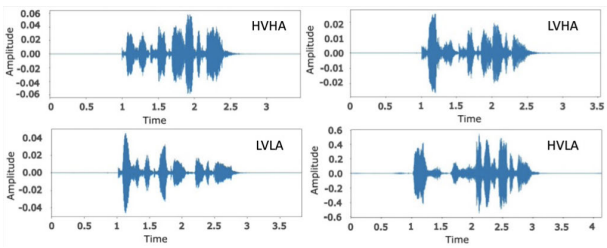


**FIGURE 6.** Time-Amplitude plots by VA classes.

Subsequently, we applied feature extraction techniques to the augmented audio data, including the Mel-frequency cepstral coefficient (MFCC), zero crossing rate (ZCR), and root mean square energy (RMSE). MFCC is a crucial technique for extracting frequency characteristics from audio data, capturing the frequency range and speech spectrum information, which is essential for analyzing speaker articulation, speech characteristics, and emotional expression. RMSE serves as an indicator of the energy level in the audio signal and provides valuable information for detecting emotional variations related to speech tone. Additionally, ZCR indicates the points in the audio signal where the sign changes, allowing analysis of changes in speech energy and tone. These features, including MFCC, ZCR, and RMSE, were then stacked to augment the data and serve as input to the model.

The dataset used in the SER consists of specific emotion labels (happy, calm, sad, angry, disgusted, fearful, surprised, and neutral). However, for this study, we mapped each emotion category into four quadrants based on VA, such as HVHA, HVLA, LVLA, and LVHA. In cases where the mapping into four quadrants was ambiguous, such as for the emotions, surprised and neutral, we substituted them with similar emotions, alert in the HVHA quadrant and serene in the LVHA quadrant, respectively. Therefore, happy and surprised were mapped to HVHA, calm and neutral to LVHA, sad to LVLH, and angry, disgusted, and fearful to HVLA.

#### 3) MODEL ARCHITECTURE

Like HR, speech data are characterized by time-series data. Therefore, when implementing the SER architecture, we combined the 1D CNN with LSTM layers which leverages the CNN's ability to extract temporal features and the LSTM's capacity to understand sequence dependencies. This architecture is particularly appropriate for handling the sequential nature of speech associated with different emotions. As indicated in the Ablation Study 8 of Table 2, by performing experimentally the alteration of model architectures between 1D CNN-Flatten, 1D CNN-LSTM, and DCNN, we constructed the 1D CNN-Flatten-based model in the SER. Figure 7 represents a block diagram of the architecture of the SER.

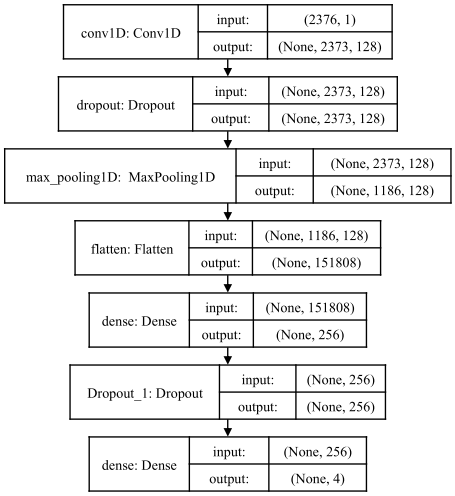


**FIGURE 7.** Block diagram of the SER: 1D CNN-Flatten.

### D. MULTIMODAL FUSION

Based on the HER and the SER constructed in the previous sections, we conducted multimodal fusion using a score-based fusion algorithm to combine softmax values obtained from each modality.

To implement score-based fusion, we first applied a grouping algorithm to create HR-speech instance groups. Since we developed the HER and the SER using different datasets, the temporal axes for each modality are independent. However, considering that humans experience the same emotion at the same time, we grouped single modality instances that

correspond to the same emotion, thereby creating HR-speech instances.

Feature-level Fusion directly combining features from each modality can lead to problems such as a sharp increase in feature vector dimensions, known as the curse of dimensionality, and may also face problems due to data discrepancies arising from different modalities having distinct features. Therefore, we opted for a score-based fusion approach when performing multimodal fusion. The score-based fusion method utilizes the prediction results, softmax probabilities, from the HER and the SER. This process involves generating these probabilities for each modality within grouped HR-speech instances and averaging them. We use three different methods for the average: A Simple Average, a Weighted Average using performances, and a Dynamic Weight Adjustment.

The simple average is to divide the values of softmax derived from the HER and the SER by the number of modalities, in our case of two, as described in

$$Final = \frac{\sum_i^N S_i}{N}, \tag{11}$$

where $N$ is the number of modalities, and $S_i$ represents the softmax of each modality $i$.

The weighted average approaches using performances and the dynamic weight adjustment are weighted averaging methods that derive the final softmax shown in

$$Final = \frac{\sum_i^N (S_i \times w_i)}{\sum w}, \tag{12}$$

where $w$ denotes the weight of modality, and $w_i$ denotes the weight of modality. These two methods differ in which weights are used. The former's weight takes into account the performances of the HER and the SER, and the latter considers the modality's quality based on the performances of the HER and the SER. The weights, considering data quality, are calculated through

$$w_i = \frac{1}{1 + \alpha U_i}, \tag{13}$$

where $\alpha$ represents a hyperparameter to control the sensitivity of weight adjustment to uncertainty, and $U_i$ represents uncertainty for modality $i$. Additionally, to calculate an Uncertainty ($U_i$), we utilize Laplace Uncertainty Calculation as described in

$$P(\theta \mid D) \approx q(\theta) = \mathcal{N}(\theta; \mu, \Sigma), \tag{14}$$

where $\theta$ is parameters (in our case, the model's confidence in predictions such as performances), $D$ represents data, $\mu$ denotes the mode of the posterior, and $\Sigma$ is the inverse of the negative Hessian of the log posterior at its mode. By applying this method, we compute quality-based weights for each modality.

As the final prediction fusion method for score-based fusion, we experimentally selected the weighted average using performances as indicated in Table 4.

## IV. RESULTS AND ANALYSIS

In this section, we present the results of the ablation studies for the HER and the SER, as well as the average method of the multimodal fusion. We also include the performances of the optimal HER, SER, and MER and the corresponding confusion matrices.

In our experiments, we converted the F1 score, which ranges between 0 to 1, to a percentage for evaluating the models' performance. The F1 score is a statistical measurement used to assess the accuracy of classification models. It is calculated by combining precision and recall, providing a balance between the two, and making it useful when dealing with imbalanced class distributions. Precision represents the ratio of samples correctly predicted as positive by the model out of all samples predicted as positive. Recall indicates the ratio of actual positive samples correctly predicted as positive by the model out of all actual positive samples. The F1 score is the harmonic mean of precision and recall, providing a high value when both are balanced. A higher F1 score implies better overall model performance, particularly when precision and recall are balanced. These performance metrics values are calculated using true positives (TP) which is the number of samples correctly predicted as positive by the model, false positives (FP) which denotes the number of samples incorrectly predicted as positive by the model, and false negatives (FN) which represents the number of samples incorrectly predicted as negative by the model among the actual positives. The equations for these evaluation metrics are shown in

$$Precision = \frac{TP}{TP + FP}, \tag{15}$$

$$Recall = \frac{TP}{TP + FN}, \tag{16}$$

$$F1\ score = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \tag{17}$$

### A. ABLATION STUDY

During our research on emotion recognition, we determined the model's configuration through experimental methods. In this section, we show the results of the ablation studies. Table 1 presents the ablation results in the HER of altering the model type, the batch size, the drop rate in the Dropout layer, the loss function, the optimizer, the HR instance size, and the oversampling method. Table 2 shows the SER results of altering the model type, the batch size, the drop rate in the Dropout layer, and the optimizer.

#### 1) OPTIMAL CONFIGURATIONS OF THE HEART RATE-BASED EMOTION RECOGNITION MODEL

For the ablation studies of the HER, we initially set the baseline configuration as follows: the epoch is 150, the batch size is 32, the drop rate is 0.5, the optimizer is adaptive moment estimation (Adam), the loss function is categorical cross-entropy, the instance size is 15 seconds (from 15 seconds before to 0 seconds), and the oversampling method is KMeansSMOTE.

**TABLE 1.** Ablation studies for the HER (Ablation Study 1-7).

(ABLATION STUDY 1-7).

**Ablation Study 1: Altering the Model Type**

| No. | Model Type | F1 score (%) | Finding |
|---|---|---|---|
| 1 | **1D CNN-Flatten** | **47.31** | Improved |
| 2 | 1D CNN-LSTM | 47.16 | Dropped |
| 3 | DCNN | 37.12 | Dropped |

**Ablation Study 2: Altering the Batch Size**

| No. | Batch Size | F1 score (%) | Finding |
|---|---|---|---|
| 1 | 16 | 41.95 | Dropped |
| 2 | **32** | **47.31** | Criteria |
| 3 | 64 | 45.00 | Dropped |
| 4 | 128 | 42.25 | Dropped |

**Ablation Study 3: Altering the Drop Rate**

| No. | Drop Rate | F1 score (%) | Finding |
|---|---|---|---|
| 1 | 0.3 | 44.99 | Dropped |
| 2 | **0.5** | **47.31** | Criteria |
| 3 | 0.7 | 40.47 | Dropped |

**Ablation Study 4: Altering the Loss Function**

| No. | Loss Function | F1 score (%) | Finding |
|---|---|---|---|
| 1 | **Categorical cross-entropy** | **47.31** | Criteria |
| 2 | Poisson | 43.32 | Dropped |
| 3 | KL Divergence | 33.44 | Dropped |

**Ablation Study 5: Altering the Optimizer**

| No. | Optimizer | F1 score (%) | Finding |
|---|---|---|---|
| 1 | AdaGrad | 37.84 | Dropped |
| 2 | **Adam** | **47.31** | Criteria |
| 3 | Nadam | 43.51 | Dropped |
| 4 | RMSProp | 40.65 | Dropped |
| 5 | SGD | 38.85 | Dropped |

**Ablation Study 6: Altering the Instance Size**

| No. | Instance Size (sec) | F1 score (%) | Finding |
|---|---|---|---|
| 1 | 15 (-15 to 0) | 47.31 | Criteria |
| 2 | 30 (-30 to 0) | 51.11 | Improved |
| 3 | 10 (-10 to 0) | 38.86 | Dropped |
| 4 | 20 (-10 to 10) | 47.41 | Improved |
| 5 | **30 (-15 to 15)** | **52.06** | Improved |
| 6 | 10 (-5 to 5) | 39.17 | Dropped |
| 7 | 30 (-20 to 10) | 51.34 | Improved |
| 8 | 15 (-10 to 5) | 44.18 | Dropped |

**Ablation Study 7: Altering the Oversampling Method**

| No. | Oversampling + Model | F1 score (%) | Finding |
|---|---|---|---|
| 1 | KmeansSMOTE + 1D CNN-Flatten | 52.06 | Criteria |
| 2 | **KmeansSMOTE + 1D CNN-LSTM** | **57.65** | Improved |
| 3 | KmeansSMOTE + DCNN | 41.43 | Dropped |
| 4 | SMOTE + 1D CNN-Flatten | 49.83 | Dropped |
| 5 | SMOTE + 1D CNN-LSTM | 53.28 | Improved |
| 6 | SMOTE + DCNN | 33.24 | Dropped |
| 7 | RandomOverSampler + 1D CNN-Flatten | 51.01 | Dropped |
| 8 | RandomOverSampler + 1D CNN-LSTM | 51.49 | Dropped |
| 9 | RandomOverSampler + DCNN | 36.68 | Dropped |

Table 1 shows the ablation studies of the HER before the fusion. The detailed description of the ablation results is given below:

- Ablation Study 1: Altering the Model Type

In this experiment, we based the study on the baseline configuration and compared the performance by altering the model architecture between 1D CNN-Flatten, 1D CNN-LSTM, and DCNN. The results showed that 1D CNN-Flatten performed the best with the F1 score of 47.31%. Therefore, we proceeded with this model type.

- Ablation Study 2: Altering the Batch Size

In Ablation Study 1, based on the model type derived, which was 1D CNN-Flatten, and the baseline parameters, we conducted experiments by varying the batch size. However, it was evident that the baseline batch size of 32 yielded the highest performance.

- Ablation Study 3: Altering the Drop Rate

Like Ablation Study 2, we kept other parameters fixed and experimented with different drop rates for the Dropout layer in the model. Once again, the baseline value of 0.5 provided the highest performance.

- Ablation Study 4: Altering the Loss Function

To study the impact of the loss function, we used various multi-categorical loss functions, including categorical cross-entropy, Poisson, and Kullback-Leibler divergence (KL Divergence). The results indicated that the baseline loss function, Categorical Cross-Entropy, achieved the best performance.

- Ablation Study 5: Altering the Optimizer

For the optimizer-based ablation study, we explored commonly used optimizers such as adaptive gradient (AdaGrad), Adam, Nesterov-accelerated adaptive moment Adam (Nadam), root mean square propagation (RMSProp), and stochastic gradient descent (SGD). Once again, the baseline optimizer, Adam, outperformed the others.

- Ablation Study 6: Altering the Instance Size

Following the determinations from Ablation Study 1-5 regarding model architecture, epoch, batch size, drop rate, loss function, and optimizer, we experimented with altering the HR instance size. We considered that users would enter their emotions after experiencing them, so we would like to consider the "before second" of the instances more. The results showed that larger instance sizes than the baseline 15 seconds led to performance improvement.

Consequently, we selected instances ranging from 15 seconds before to 15 seconds after, which exhibited the optimal performance of 52.06%.

• Ablation Study 7: Altering the Oversampling Method

After establishing the model configuration through Ablation Study 1-6, we conducted Ablation Study 7 to explore the optimal combination of an oversampling method and model type for mitigating class imbalance. While keeping the configuration determined through Ablation Study 2-6 fixed, we experimented with different oversampling methods and models. Interestingly, the 1D CNN-LSTM model showed improved performance, unlike Ablation Study 1 where the 1D CNN-Flatten model performed the best. This experiment confirmed that the combination of KMeansSMOTE and 1D CNN-LSTM achieved the highest performance, reaching 57.65%, and was thus chosen as the optimal model.

### 2) OPTIMAL CONFIGURATIONS OF THE SPEECH-BASED EMOTION RECOGNITION MODEL

For the SER, we established baseline configuration as follows: the epoch is 50, the batch size is 64, the drop rate is 0.5, the optimizer is Adam, and the loss function is Categorical Cross-Entropy.

Table 2 shows the ablation studies of the SER before the fusion. The detailed description of the ablation results is given below:

• Ablation Study 8: Altering the Model Type

In the SER model, we used the identical model types as in the HER. In Ablation Study 8, hence, we compared the performance by varying the model architecture between 1D CNN-Flatten, 1D CNN-LSTM, and DCNN. As a result, the 1D CNN-Flatten shows the best performance of 68.47%, as the optimal model.

• Ablation Study 9: Altering the Batch Size

With 1D CNN-Flatten and baseline parameters as a foundation, we conducted experiments by changing the batch size. The results showed that as the batch size decreased, the classification performance improved. Ultimately, reducing the batch size from 64 to 16 yielded the highest performance of 72.11%, and we proceeded with this configuration.

• Ablation Study 10: Altering the Drop Rate

To determine the optimal hyperparameter for the model's Dropout layer, Ablation study 10 was performed. When comparing the F1 score with the drop rate of 0.3, 0.5, and 0.7, the drop rate of 0.3 demonstrated the significant highest performance of 77.72%.

• Ablation Study 11: Altering the Loss Function

Keeping the configuration derived from Ablation study 8-10 unchanged, we experimented with different loss functions. The loss functions were the same as those used in the HER ablation study. Like Ablation study 10, it was confirmed that the baseline loss function was the most optimal.

• Ablation Study 12: Altering the Optimizer

To determine the model's optimizer, we conducted experiments by switching between AdaGrad, Adam, Nadam, RMSProp, and SGD optimizers. During this experimentation, the SGD optimizer, with a performance of 80.38%, outperformed the existing Adam optimizer. Therefore, we selected SGD as the final optimizer.

**TABLE 2. Ablation studies for the SER (Ablation Study 8-12).**

(ABLATION STUDY 8-12).

**Ablation Study 8: Altering the Model Type**

| No. | Model Type | F1 score (%) | Finding |
|---|---|---|---|
| 1 | **1D CNN-Flatten** | **68.47** | Criteria |
| 2 | 1D CNN-LSTM | 20.70 | Dropped |
| 3 | DCNN | 58.67 | Dropped |

**Ablation Study 9: Altering the Batch Size**

| No. | Batch Size | F1 score (%) | Finding |
|---|---|---|---|
| 1 | **16** | **72.11** | Improved |
| 2 | 32 | 69.76 | Improved |
| 3 | 64 | 68.47 | Criteria |
| 4 | 128 | 64.42 | Dropped |

**Ablation Study 10: Altering the Drop Rate**

| No. | Drop Rate | F1 score (%) | Finding |
|---|---|---|---|
| 1 | **0.3** | **77.72** | Improved |
| 2 | 0.5 | 72.11 | Criteria |
| 3 | 0.7 | 61.81 | Dropped |

**Ablation Study 11: Altering the Loss Function**

| No. | Loss Function | F1 score (%) | Finding |
|---|---|---|---|
| 1 | **Categorical cross-entropy** | **77.72** | Criteria |
| 2 | Poisson | 48.28 | Dropped |
| 3 | KL Divergence | 48.28 | Dropped |

**Ablation Study 12: Altering the Optimizer**

| No. | Model Type | F1 score (%) | Finding |
|---|---|---|---|
| 1 | AdaGrad | 64.28 | Dropped |
| 2 | Adam | 77.72 | Criteria |
| 3 | Nadam | 72.61 | Dropped |
| 4 | RMSProp | 57.92 | Dropped |
| 5 | **SGD** | **80.38** | Improved |

### 3) OPTIMAL FUSION METHOD OF THE MULTIMODAL EMOTION RECOGNITION MODEL

After conducting these 12 ablation studies, the optimal configurations for the HER and the SER are compared in Table 3.

Using the optimal configurations derived from the previous Ablation study 1-12, we performed the score-based fusion. When conducting the multimodal fusion, we combined the predicted softmax values from each model using three different methods: the simple averaging method, the weighted average method that utilizes the normalized performance of each model, 0.5765 and 0.8038, as the weights, and the dynamic weight adjustment method, which sets weights based on an evaluation of data quality. In the performance

**TABLE 3.** Configurations of the emotion recognition models.

| Configuration | HER | SER |
|---|---|---|
| **Model Structure** | 1D CNN-LSTM | 1D CNN-Flatten |
| **Oversampling** | KmeansSMOTE | - |
| **Instance Size (seconds)** | 30 (-15s to 15s) | 2.5 (offset: 0.6s) |
| **Optimizer** | Adam | SGD |
| **Loss Function** | Categorical cross-entropy | Categorical cross-entropy |
| **Epoch** | 150 | 50 |
| **Batch Size** | 32 | 16 |
| **Drop Rate** | 0.5 | 0.3 |

comparison based on the fusion method presented in Table 4, the weighted average method using the model performances showed the highest performance of 84.22%.

**TABLE 4.** Ablation study of the multimodal fusion (Ablation Study 13).

| Ablation Study 13: Altering the Fusion Method | | | |
|---|---|---|---|
| **No.** | **Fusion Method** | **Weight [HR, Speech]** | **F1 score (%)** |
| 1 | Simple Average | [0.5000, 0.5000] | 83.50 |
| 2 | Weighted Average using performances | [0.4203, 0.5797] | **84.22** |
| 3 | Dynamic Weight Adjustment | [0.4999, 0.5001] | 83.50 |

### B. EVALUATION METRICS

Table 5 and Figure 8 represent the F1 scores and confusion matrices for the HER, the SER, and the MER, as determined through the ablation study in the preceding sections. As seen in Table 5, the HER achieved a performance of 57.65%, the SER reached 80.38%, and the MER attained 84.22%, providing that the MER outperforms the others. This demonstrates that the integration of HR and speech data provides a more holistic understanding of emotional states. Additionally, when unable to obtain one of the data sources, the model can still perform emotion recognition tasks, making it useful in realistic scenarios.

**TABLE 5.** F1 score of the emotion recognition models.

| Model | F1 score (%) |
|---|---|
| **HER** | 57.65 |
| **SER** | 80.38 |
| **MER** | 84.22 |

## V. DISCUSSION

In this section, we compare our approach with the latest models and discuss its advantages and limitations. The comparison is made across physiological data, speech, and multimodal emotion recognition, as represented in Table 6.

**TABLE 6.** Comparison with the latest models.

| Modality | Dataset | Method | Class | Performance | References |
|---|---|---|---|---|---|
| PPG | MANNOB-HCI | 1D CNN-LSTM | 2 (VA) | Accuracy of 73.33% in valence, 60.00% in arousal | [5] |
| HR | TERS-ER, Ulm-TSST | 1D CNN-LSTM | 4 (VA quadrant) | F1 score of 57.65% | This study |
| Speech | IEMOCAP | Hybrid Attention Networks | 4 (happy, sad, neutral, angry) | F1 score of 56.06% | [13] |
| Speech | CREMA-D, TESS, RAVDESS, SAVEE | 1D CNN-Flatten | 4 (VA quadrant) | F1 score of 80.38% | This study |
| BVP, EDA, ST, Facial Expressions | Acquisition from hearing-impaired children | Feature-level Fusion based on MTCNN | 3 (positive, negative, neutral) | Accuracy of 74.96% | [29] |
| BVP, EDA, ST, Facial Expression, Speech, Keystroke | User study | Attention-based LSTM | 2 (positive, negative) | Accuracy of 89.2% | [20] |
| HR, Speech | HR: TERS-ER, Ulm-TSST Speech: CREMA-D, TESS, RAVDESS, SAVEE | Score-based Fusion based on 1D CNN-LSTM and 1D CNN-Flatten | 4 (VA quadrant) | F1 score of 84.22% | This study |

Regarding emotion recognition using physiological sensor data, Mellouk and Handouzi [5] achieved 73.33% for valence and 60.00% for arousal using PPG signals. Although this study demonstrated commendable results with unique data, its practicality in daily-life scenarios may be limited due to the computational expenses and dataset complexity associated with PPG data. Additionally, their study focused on a smaller set of emotional classes compared to our approach. Concerning emotion recognition using speech data sources, the study by Zhang et al. [13] achieved an F1 score of 56.06% using hybrid attention networks for four emotional classes. Notably, despite the use of hybrid attention networks, our approach achieved higher performances for the same number of emotional categories. For multimodal emotion recognition, Takir et al. [29] achieved a recognition rate of 74.96% using physiological signals, including Blood Volume Pulse (BVP), Electro Dermal Activity (EDA), and Skin Temperature (ST), and facial expressions from hearting-impaired children for three emotional categories. Similarly, Yang et al. [20] attained 89.2% accuracy using multiple physiological sensor data sources, including BVP, EDA, and ST, facial expression, speech, and keystroke. Although these studies not only conducted their studies with unique data sources and more modalities but also achieved impressive accuracies, they considered smaller emotional categories than ours.

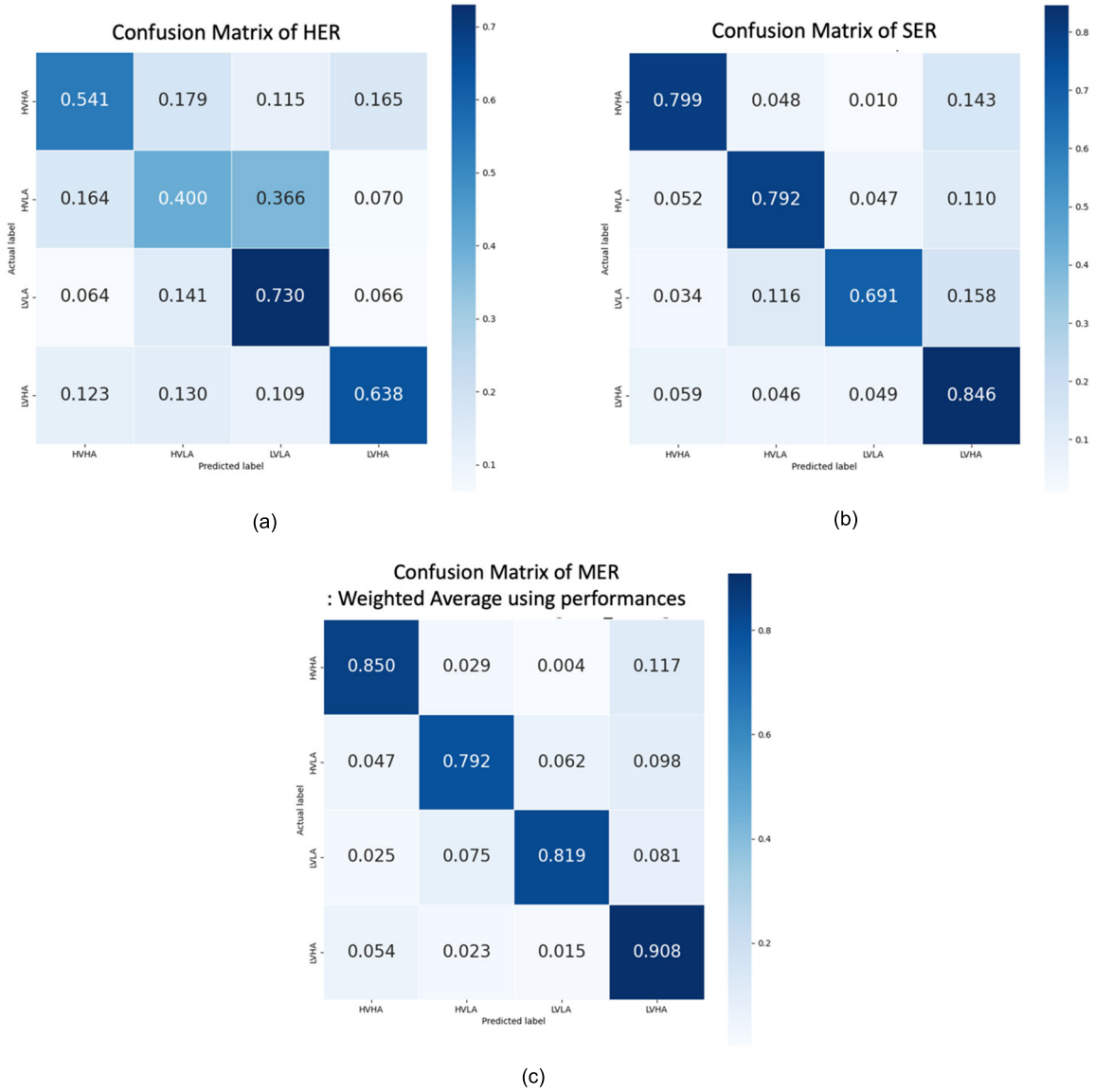


**FIGURE 8.** Confusion matrices of emotion recognition models: a) HER; b) SER; C) MER.

Despite the notable state-of-the-art models with their impressive accuracies and unique approaches, our study offers a balance between performance and feasibility in real-world applications. Our approach achieves an overall emotion recognition accuracy of 84.22% by leveraging data sources easily collectible from low-cost wearable devices like smartwatches and employing the fusion approach that minimizes computational complexity.

Firstly, our system offers enhanced portability and affordability as it utilizes emotion-related data sources such as HR and speech, which can be collected using only low-cost smartwatches. This not only improves the practicality of emotion recognition systems but also offer convenience without necessitating the wearing or usage of expensive or cumbersome devices. With no constraints on daily life, our system is well-suited for emotion-tracking applications in everyday scenarios, allowing for seamless integration into users' routines.

Moreover, our system is designed to be lightweight, emphasizing efficiency and simplicity in its design. By adopting the score-based fusion approach for integrating HR and speech modalities, we eliminate the need for additional neural networks and reduce computational demands. This lightweight architecture allows for the embedding of our system in low-cost wearables like smartwatches. Despite its streamlined architecture, our approach maintains robust performance in emotion recognition accuracy, ensuring its compatibility and usability in real-world scenarios.

Overall, the portability and simplicity of our proposed approach contribute to its effectiveness in diverse settings, making it suitable for practical applications such as stress management and health monitoring.

Despite these advantages of our approach, there remain limitations to consider. While our MER achieved an accuracy of 84.22%, the HER achieved a comparatively lower accuracy of 57.65% when compared to the SER with 80.38%. This performance variation may indicate potential challenges in achieving consistent accuracy across different modalities and datasets.

Our approach prioritized the practicality of emotion tracking by leveraging the simplicity and convenience of using low-cost smartwatches, which inherently restricts the available modalities to those feasible for collection by these devices. Consequently, we focused on utilizing HR data, despite its relatively lower performance. This disparity underscores the challenge of effectively utilizing HR data for emotion recognition, particularly within the practical constraints of monitoring emotions using only smartwatches in users' daily lives.

To address this limitation, future research endeavors could explore alternative approaches, such as incorporating additional physiological sensor modalities like EEG and ST, should they become accessible through advancements in low-cost wearable technology. By integrating these modalities while maintaining a focus on practicality and user-friendly design, our system could further enhance its effectiveness in real-world emotion-tracking scenarios.

## VI. CONCLUSION

We propose the MER system as a significant contribution to practical affective computing research. Our approach focuses on emotion tracking by leveraging emotion-related data sources, such as HR and speech, easily obtainable from low-cost smartwatches in users' daily lives. The continuous monitoring of HR, which can be done seamlessly without needing any extra effort from the user, combined with the distinct qualities of speech expressions, allows for a thorough analysis of human emotional states. In response, our approach implements and integrates HER and SER with lightweight architectures.

In the context of the individual model, HER and SER utilize deep learning for discerning crucial patterns. The ablation study reveals a 57.65% accuracy for the 1D CNN-LSTM-based HER model and an 80.38% accuracy for the 1D CNN-Flatten-based SER model. Our system's portability and affordability enhance the practicality of emotion recognition systems without requiring expensive or cumbersome devices.

In the context of our fusion method, employing the score-based weighted average using models' performances, achieves an 84.22% performance for MER. This outcome underscores the significance of utilizing multiple modalities in emotion recognition, not only for a comprehensive understanding of the human emotional state but also for ensuring robustness in real-world scenarios with numerous variables. Moreover, our system's lightweight architecture, achieved without the need for additional neural networks, demonstrates efficiency in operation. Additionally, our multimodal fusion approach is easily adaptable to include additional modalities, maintaining robust precision even in scenarios where not all modalities are available, thereby enhancing its versatility and practicality.

Building on the findings of our MER system, there are several promising avenues for future research. Immediate extensions could involve incorporating additional modalities such as EEG and ST for enhanced emotion recognition by leveraging low-cost wearables. As advancements progress in on-device systems for lightweight deep learning-based models, low power consumption, and rapid responses, including neural processing units, our study could expand to embed in chips for real-time as well as energy-efficient emotion management applications. Additionally, it would be intriguing to explore applications that not only track users' emotional states but also contribute to stress reduction, such as music recommendation and conversational robot systems.

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**EESUN MOON** received the B.S. degree in intelligent mechatronics engineering and data science from Sejong University, Seoul, Republic of Korea.

From September 2021 to 2023, she was a Research Assistant with the Mobile Intelligent Embedded Systems Laboratory, Sejong University. Her research interests include artificial intelligence, natural language processing, human–robot interaction, and affective computing.

**A. S. M. SHARIFUZZAMAN SAGAR** (Student Member, IEEE) received the bachelor's degree from China University of Mining and Technology, China, in 2020, and the master's degree from Sejong University, Seoul, Republic of Korea, in 2022.

He is currently a Research Assistant with the Department of Artificial Intelligence and Robotics, Sejong University. His current research interests include deep learning, embedded AI systems, the Internet of Things, and human–computer interaction. He is also a Non-Voting Member of the IEEE Sensors Council Standards Committee.

**HYUNG SEOK KIM** (Member, IEEE) received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in electrical engineering and computer engineering from Seoul National University, Republic of Korea, in 1996, 1998, and 2004, respectively. In 2002 and 2003, he was a Research Assistant with the University of Virginia. From 2004 to 2006, he was with Samsung Electronics. He is currently a Faculty Member with the Department of Artificial Intelligence and Robotics, Sejong University. His research interests include the Internet of Things and AI robot systems.